Status	Finished
Started	Monday, 3 August 2026, 4:09 PM
Completed	Monday, 3 August 2026, 4:35 PM
Duration	26 mins 27 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Write a program to find whether the given input number is Even.

If the given number is even, the function should return 2 else it should return 1.

Note: The number passed to the program can either be negative, positive or zero. Zero should be treated as Even.

For example:

Input	Result
100	2
1001	1

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all lines from standard input automatically
    for line in sys.stdin:
        # Skip empty lines
        if not line.strip():
            continue

        try:
            # Convert the input line to an integer
            number = int(line.strip())

            # If even, return/print 2, else 1
            if number % 2 == 0:
                print(2)
            else:
                print(1)

        except ValueError:
            # Safely handle any non-integer unexpected inputs
            continue

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	100	2	2	✓
✓	1001	1	1	✓
✓	0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Note:

Dont use if-else. Operators alone must be used .

A team from the Rotract club had planned to conduct a rally to create awareness among the Coimbatore people to donate blood. They conducted the rally successfully. Many of the Coimbatore people realized it and came forward to donate their blood to nearby blood banks. The eligibility criteria for donating blood are people should be above or equal to 18 and his/ her weight should be above 40. There was a huge crowd and staff in the blood bank found it difficult to manage the crowd. So they decided to keep a system and ask the people to enter their age and weight in the system. If a person is eligible he/she will be allowed inside.

Write a program and feed it to the system to find whether a person is eligible or not.

Input Format:

Input consists of two integers that correspond to the age and weight of a person respectively.

Output Format:

Display True(IF ELIGIBLE)

Display False (if not eligible)

Sample Input

19

45

Sample Output

True

For example:

Input	Result
18 40	False

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all inputs from standard input
    input_data = sys.stdin.read().split()

    # If no inputs are found, stop the program safely
    if not input_data:
        return

    # Assign age and weight from the input array
    age = int(input_data[0])
    weight = int(input_data[1])

    # The eligibility logic using strictly operators (No if-else)
    # Age must be greater than or equal to 18 AND weight must be strictly greater than 40
    is_eligible = (age >= 18) and (weight > 40)

    # Directly print the resulting Boolean value
    print(is_eligible)

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	19 45	True	True	✓
✓	18 40	False	False	✓
✓	18 42	True	True	✓
✓	16 45	False	False	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Mr.Ram has been given a problem kindly help him to solve it. The input of the program is either 0 or 1. IF 0 is the input he should display "C" if 1 is the input it should display "D".There is a constraint that Mr. Ram should use either logical operators or arithmetic operators to solve the problem, not anything else.

Hint:

Use ASCII values of C and D.

Input Format:

An integer x, 0<=x<=1. .

Output Format:

output a single character "C" or "D"depending on the value of x.

```
Input 1:
0
Output 1:
C
```

```
Input 2:
1

Output 1:
D
```

For example:

Input	Result
0	C

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read the integer input from standard input
    x = int(sys.stdin.read().strip())

    # Calculate the ASCII value using arithmetic addition
    # If x is 0: 67 + 0 = 67 ('C')
    # If x is 1: 67 + 1 = 68 ('D')
    ascii_value = 67 + x

    # Convert the ASCII number back to a character and print
    print(chr(ascii_value))

if __name__ == "__main__":
    main()
```

	Input	Expected	Got	
✓	0	C	C	✓
✓	1	D	D	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

In the 1800s, the battle of Troy was led by Hercules. He was a superstitious person. He believed that his crew can win the battle only if the total count of the weapons in hand is in multiple of 3 and the soldiers are in an even number of count. Given the total number of weapons and the soldier's count, Find whether the battle can be won or not according to Hercules's belief. If the battle can be won print True otherwise print False.

Input format:

Line 1 has the total number of weapons

Line 2 has the total number of Soldiers.

Output Format:

If the battle can be won print True otherwise print False.

Sample Input:

32

43

Sample Output:'

False

For example:

Input	Result
32 43	False

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all inputs from standard input
    input_data = sys.stdin.read().split()

    if not input_data:
        return

    # Assign weapons and soldiers from the input stream
    weapons = int(input_data[0])
    soldiers = int(input_data[1])

    # Condition 1: Weapons must be a multiple of 3 (remainder is 0)
    # Condition 2: Soldiers must be an even number (remainder is 0)
    can_win = (weapons % 3 == 0) and (soldiers % 2 == 0)

    # Directly print the resulting Boolean value (True or False)
    print(can_win)

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	32 43	False	False	✓
✓	273 7890	True	True	✓

	Input	Expected	Got	
✓	800 4590	False	False	✓
✓	6789 32996	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a program that returns the last digit of the given number. Last digit is being referred to the least significant digit i.e. the digit in the ones (units) place in the given number.

The last digit should be returned as a positive number.

For example,

if the given number is 197, the last digit is 7

if the given number is -197, the last digit is 7

For example:

Input	Result
197	7
-197	7

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all input from standard input
    input_data = sys.stdin.read().split()

    for item in input_data:
        # Convert input to integer
        number = int(item)

        # 1. abs(number) converts negative numbers to positive
        # 2. % 10 extracts the units place digit
        last_digit = abs(number) % 10

        # Print the positive last digit
        print(last_digit)

if __name__ == "__main__":
    main()
```

	Input	Expected	Got	
✓	197	7	7	✓
✓	-197	7	7	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Complete the program to convert days into years, month and days. (Ignoring leap year and considering 1 month is 30 days)

Sample Test Cases

Test Case 1

Input

375

Output

YEARS: 1 MONTH: 0 DAYS: 10

Test Case 2

Input

200

Output

YEARS: 0 MONTH: 6 DAYS: 20

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read the input days from standard input
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    total_days = int(input_data[0])

    # 1. Calculate total full years using integer division
    years = total_days // 365

    # 2. Get the remaining days after extracting full years
    remaining_days_after_years = total_days % 365

    # 3. Calculate full months from the leftover days
    months = remaining_days_after_years // 30

    # 4. Get the final leftover days after extracting full months
    days = remaining_days_after_years % 30

    # Print the formatted output exactly matching the sample test cases
    print(f"YEARS: {years} MONTH: {months} DAYS: {days}")

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	375	YEARS: 1 MONTH: 0 DAYS: 10	YEARS: 1 MONTH: 0 DAYS: 10	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a python program that takes a integer between 0 and 15 as input and displays the number of '1' s in its binary form.(Hint:use python bitwise operator.

Sample Input

3

Sample Output:

2

Explanation:

The binary representation of 3 is 011, hence there are 2 ones in it. so the output is 2.

For example:

Input	Result
3	2

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read the input integer from standard input
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    number = int(input_data[0])
    ones_count = 0

    # Process the number bit by bit using bitwise operators
    # Loop works as long as there are bits remaining to process
    while number > 0:
        # number & 1 checks if the rightmost bit is a 1
        ones_count += number & 1

        # number >> 1 shifts all bits to the right by 1 position
        number = number >> 1

    print(ones_count)

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	3	2	2	✓
✓	5	2	2	✓
✓	15	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

An online retailer sells two products: widgets and gizmos. Each widget weighs 75 grams. Each gizmo weighs 112 grams. Write a program that reads the number of widgets and the number of gizmos from the user. Then your program should compute and display the total weight of the parts.

Sample Input:

10

20

Sample Output:

The total weight of all these widgets and gizmos is 2990 grams.

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all inputs from standard input as a list of strings
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Access the list elements by their indices
    widgets = int(input_data[0])
    gizmos = int(input_data[1])

    # Calculate the total weight based on product specifications
    total_weight = (widgets * 75) + (gizmos * 112)

    # Display the final output precisely matching the required string format
    print(f"The total weight of all these widgets and gizmos is {total_weight} grams.")

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	10 20	The total weight of all these widgets and gizmos is 2990 grams.	The total weight of all these widgets and gizmos is 2990 grams.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Rohit wants to add the last digits of two given numbers.

For example,

If the given numbers are 267 and 154, the output should be 11.

Below is the explanation:

Last digit of the 267 is 7

Last digit of the 154 is 4

Sum of 7 and 4 = 11

Write a program to help Rohit achieve this for any given two numbers.

Note: Tile sign of the input numbers should be ignored.

i.e.

if the input numbers are 267 and 154, the sum of last two digits should be 11

if the input numbers are 267 and -154, the slim of last two digits should be 11

if the input numbers are -267 and 154, the sum of last two digits should be 11

if the input numbers are -267 and -154, the sum of last two digits should be 11

For example:

Input	Result
267 154	11
267 -154	11

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all inputs from standard input as a list of strings
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Extract the two numbers using their list indices
    num1 = int(input_data[0])
    num2 = int(input_data[1])

    # 1. abs() removes any negative signs
    # 2. % 10 extracts the units place digit
    last_digit1 = abs(num1) % 10
    last_digit2 = abs(num2) % 10

    # Calculate the sum of the two digits
    total_sum = last_digit1 + last_digit2

    # Print the final result
    print(total_sum)

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	267 154	11	11	✓
✓	267 -154	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Write a simple python program to find the square root of a given floating point number. The output should be displayed with 3 decimal places.

Sample Input:

8.00

Sample Output:

2.828

For example:

Input	Result
8.00	2.828
14.00	3.742

Answer: (penalty regime: 0 %)

```
import sys

def main():
    # Read all inputs from standard input as a list of strings
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Convert the first input element into a floating-point number
    number = float(input_data[0])

    # Calculate the square root using the exponentiation operator
    square_root = number ** 0.5

    # Print the result formatted to exactly 3 decimal places
    print(f"{square_root:.3f}")

if __name__ == "__main__":
    main()

```

	Input	Expected	Got	
✓	8.00	2.828	2.828	✓
✓	14.00	3.742	3.742	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.